DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-300-03 — Internal inspection of horizontal drum D-101

Area	A300 Exchangers and drum	**Task**	VI Internal inspection
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	**D-101** — D-101 horizontal vessel 60" ID × 180" — E 84'-0" N -25'-0" EL 107'-6"		
Equipment	D-101	**Lines**	6"-P300-P-1012, 4"-P300-P-1013, 1"-P150-V-6016, 1"-P150-D-6017, 1"-P150-N-3014, 3/4"-P300-I-6018, 3/4"-P300-I-6019, 3/4"-P300-SP-6020
Isometrics	ISO-6"-P300-P-1012, ISO-4"-P300-P-1013, ISO-1"-P150-V-6016, ISO-1"-P150-D-6017, ISO-1"-P150-N-3014, ISO-3/4"-P300-I-6018, ISO-3/4"-P300-I-6019, ISO-3/4"-P300-SP-6020		
Permits	General work permit, Confined-space entry permit		

Reason

Scheduled internal inspection of the drum.

Scope of work

1. Isolate D-101 per the isolation list; drain, vent, purge and gas-test for entry.
2. Open for entry; clean as needed for inspection.
3. Internal visual examination; ultrasonic thickness at the 14 TMLs.
4. Close with new gaskets on every opened joint; remove blinds; restore the line-up.

Affected items: D-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6, SW-WO-7

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
D-101	equipment	D-101 horizontal vessel 60" ID × 180"		E 84'-0" N -25'-0" EL 107'-6"
6"-P300-P-1012/B2	item	VALVE, GATE, WEDGE, OS&Y 6" P300	6"-P300-P-1012	E 86'-6" N -25'-0" EL 111'-3-7/16"
D-101-N1	nozzle	Nozzle D-101-N1 (inlet, 6" P300) on D-101		E 86'-6" N -25'-0" EL 110'-8-5/16"
4"-P300-P-1013/A2	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1013	E 98'-0" N -25'-0" EL 103'-8-3/16"
D-101-N2	nozzle	Nozzle D-101-N2 (outlet, 4" P300) on D-101		E 98'-0" N -25'-0" EL 104'-3-15/16"
3"-P300-RV-4001/2	item	VALVE, PRESSURE SAFETY VALVE 3" P300 INLET × 4" P150 OUTLET	3"-P300-RV-4001	E 94'-0" N -25'-0" EL 111'-6"
1"-P150-V-6016/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-V-6016	E 89'-0" N -25'-0" EL 111'-1/2"
D-101-N4	nozzle	Nozzle D-101-N4 (vent, 1" P150) on D-101		E 89'-0" N -25'-0" EL 110'-5-11/16"
1"-P150-D-6017/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-D-6017	E 85'-6" N -25'-0" EL 103'-11-1/2"
D-101-N5	nozzle	Nozzle D-101-N5 (drain, 1" P150) on D-101		E 85'-6" N -25'-0" EL 104'-6-3/8"
1"-P150-N-3014/B2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-N-3014	E 91'-6" N -25'-0" EL 111'-1/2"
D-101-N6	nozzle	Nozzle D-101-N6 (nitrogen, 1" P150) on D-101		E 91'-6" N -25'-0" EL 110'-5-11/16"
3/4"-P300-I-6018/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-I-6018	E 91'-6" N -28'-6-1/4" EL 107'-6"
D-101-N7	nozzle	Nozzle D-101-N7 (instrument, 0.75" P300) on D-101		E 91'-6" N -27'-11-1/2" EL 107'-6"
3/4"-P300-I-6019/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-I-6019	E 95'-0" N -28'-6-1/4" EL 107'-6"
D-101-N8	nozzle	Nozzle D-101-N8 (instrument, 0.75" P300) on D-101		E 95'-0" N -27'-11-1/2" EL 107'-6"
3/4"-P300-SP-6020/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-SP-6020	E 93'-0" N -25'-0" EL 103'-11-3/4"
D-101-N9	nozzle	Nozzle D-101-N9 (sample, 0.75" P300) on D-101		E 93'-0" N -25'-0" EL 104'-6-1/2"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	6"-P300-P-1012/B2	6"-P300-P-1012	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1012/B2) on D-101-N1	E 86'-6" N -25'-0" EL 111'-3-7/16"		
2	blind	D-101-N1	6"-P300-P-1012	Fit a blind between 6"-P300-P-1012/B2 and the nozzle flange D-101-N1	E 86'-6" N -25'-0" EL 110'-8-5/16"		

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
3	valve	4"-P300-P-1013/A2	4"-P300-P-1013	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1013/A2) on D-101-N2	E 98'-0" N -25'-0" EL 103'-8-3/16"		
4	blind	D-101-N2	4"-P300-P-1013	Fit a blind between 4"-P300-P-1013/A2 and the nozzle flange D-101-N2	E 98'-0" N -25'-0" EL 104'-3-15/16"		
5	removal	3"-P300-RV-4001/2	3"-P300-RV-4001	Remove the relief valve on D-101-N3 and blind the nozzle			
6	valve	1"-P150-V-6016/2	1"-P150-V-6016	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6016/2) on D-101-N4	E 89'-0" N -25'-0" EL 111'-1/2"		
7	blind	D-101-N4	1"-P150-V-6016	Fit a blind between 1"-P150-V-6016/2 and the nozzle flange D-101-N4	E 89'-0" N -25'-0" EL 110'-5-11/16"		
8	valve	1"-P150-D-6017/2	1"-P150-D-6017	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6017/2) on D-101-N5	E 85'-6" N -25'-0" EL 103'-11-1/2"		
9	blind	D-101-N5	1"-P150-D-6017	Fit a blind between 1"-P150-D-6017/2 and the nozzle flange D-101-N5	E 85'-6" N -25'-0" EL 104'-6-3/8"		
10	valve	1"-P150-N-3014/B2	1"-P150-N-3014	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-N-3014/B2) on D-101-N6	E 91'-6" N -25'-0" EL 111'-1/2"		
11	blind	D-101-N6	1"-P150-N-3014	Fit a blind between 1"-P150-N-3014/B2 and the nozzle flange D-101-N6	E 91'-6" N -25'-0" EL 110'-5-11/16"		
12	valve	3/4"-P300-I-6018/2	3/4"-P300-I-6018	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6018/2) on D-101-N7	E 91'-6" N -28'-6-1/4" EL 107'-6"		
13	blind	D-101-N7	3/4"-P300-I-6018	Fit a blind between 3/4"-P300-I-6018/2 and the nozzle flange D-101-N7	E 91'-6" N -27'-11-1/2" EL 107'-6"		
14	valve	3/4"-P300-I-6019/2	3/4"-P300-I-6019	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6019/2) on D-101-N8	E 95'-0" N -28'-6-1/4" EL 107'-6"		
15	blind	D-101-N8	3/4"-P300-I-6019	Fit a blind between 3/4"-P300-I-6019/2 and the nozzle flange D-101-N8	E 95'-0" N -27'-11-1/2" EL 107'-6"		
16	valve	3/4"-P300-SP-6020/2	3/4"-P300-SP-6020	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-SP-6020/2) on D-101-N9	E 93'-0" N -25'-0" EL 103'-11-3/4"		
17	blind	D-101-N9	3/4"-P300-SP-6020	Fit a blind between 3/4"-P300-SP-6020/2 and the nozzle flange D-101-N9	E 93'-0" N -25'-0" EL 104'-6-1/2"		

Inspections

Id	Inspection	Criteria	Basis
WO-300-03-I1	Internal visual examination of the shell, heads and nozzles	Findings recorded against the owner's integrity criteria	SW-NDE-4
WO-300-03-I2	Ultrasonic thickness at the 14 TMLs listed	Each reading above the owner's minimum wall	SW-WO-6

Thickness monitoring locations (SW-WO-6)

TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-300-03-TML01	D-101	Nozzle neck D-101-N1	E 86'-6" N -25'-0" EL 110'-8-5/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-03-TML02	D-101	Nozzle neck D-101-N2	E 98'-0" N -25'-0" EL 104'-3-15/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-03-TML03	D-101	Nozzle neck D-101-N3	E 94'-0" N -25'-0" EL 110'-10-5/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-03-TML04	D-101	Nozzle neck D-101-N4	E 89'-0" N -25'-0" EL 110'-5-11/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-03-TML05	D-101	Nozzle neck D-101-N5	E 85'-6" N -25'-0" EL 104'-6-3/8"		top: ____	side: ____	bottom: ____	side: ____
WO-300-03-TML06	D-101	Nozzle neck D-101-N6	E 91'-6" N -25'-0" EL 110'-5-11/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-03-TML07	D-101	Nozzle neck D-101-N7	E 91'-6" N -27'-11-1/2" EL 107'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-300-03-TML08	D-101	Nozzle neck D-101-N8	E 95'-0" N -27'-11-1/2" EL 107'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-300-03-TML09	D-101	Nozzle neck D-101-N9	E 93'-0" N -25'-0" EL 104'-6-1/2"		top: ____	side: ____	bottom: ____	side: ____
WO-300-03-TML10	D-101	Left head centre	E 82'-8-3/4" N -25'-0" EL 107'-6"		centre: ____			
WO-300-03-TML11	D-101	Right head centre	E 100'-3-1/4" N -25'-0" EL 107'-6"		centre: ____			
WO-300-03-TML12	D-101	Shell at 0.000" from the left tangent	E 84'-0" N -25'-0" EL 107'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-300-03-TML13	D-101	Shell at 6'-0" from the left tangent	E 90'-0" N -25'-0" EL 107'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-300-03-TML14	D-101	Shell at 12'-0" from the left tangent	E 96'-0" N -25'-0" EL 107'-6"		0°: ____	90°: ____	180°: ____	270°: ____

Test — Service leak test

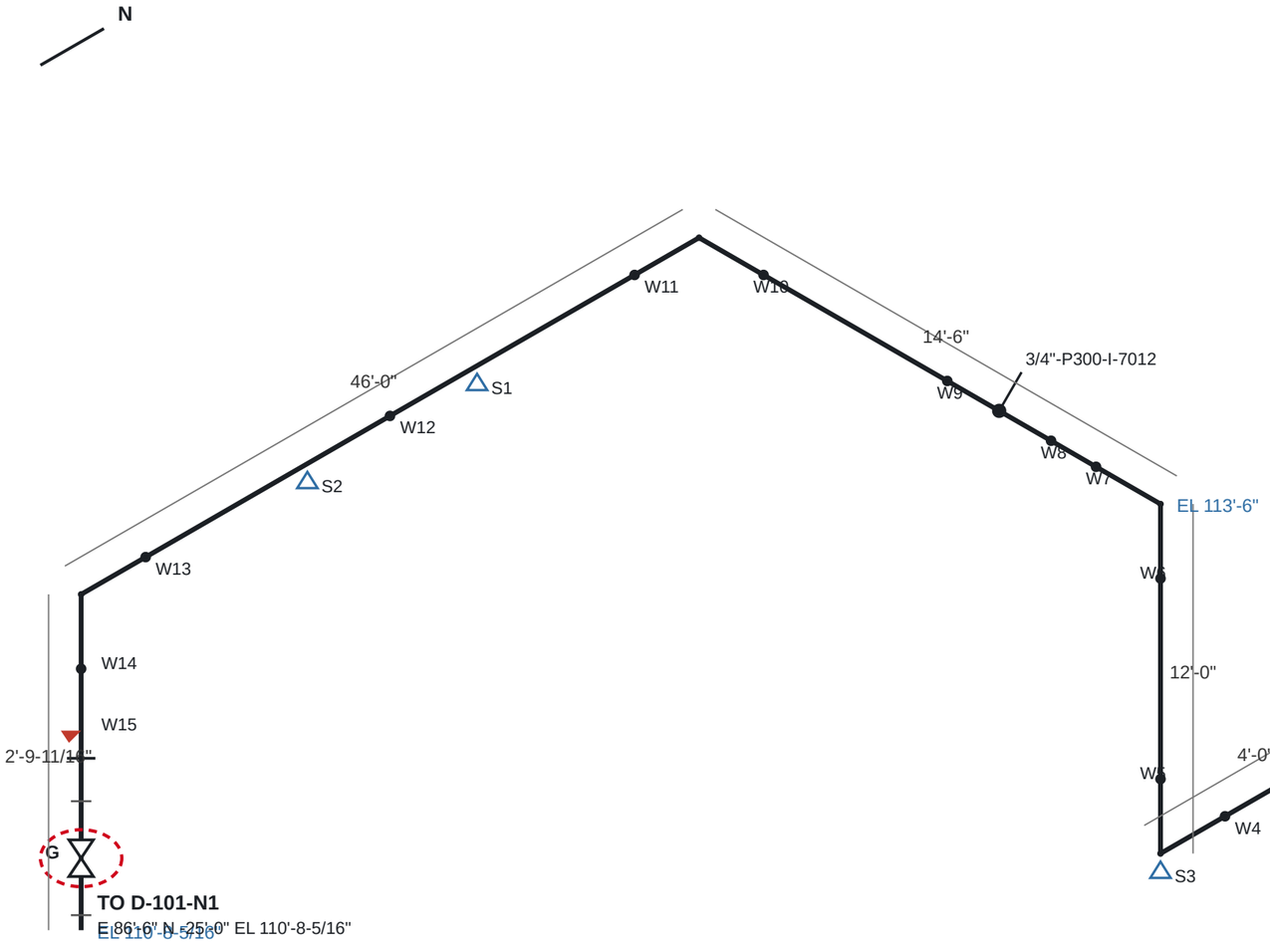
Description	Every opened joint checked at operating pressure on start-up
Basis	SW-TEST-2

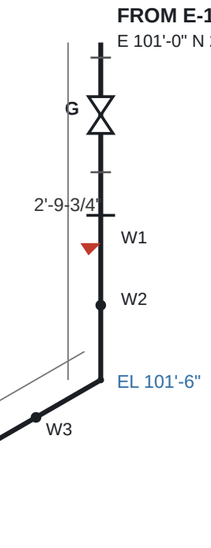
Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Equipment vented, purged and gas-tested for entry; confined-space entry permit issued	Owner entry procedure	Atmosphere safe for entry per owner limits	Gas test record, entry permit	H	H	—				
3	Opened for entry; cleaned for inspection	Owner procedure	Surfaces clean enough to inspect	Daily report	S	S	—				
4	Internal visual examination	SW-NDE-4	Findings recorded	Inspection report with photographs	W	H	W				
5	Ultrasonic thickness at 14 TMLs	SW-WO-6	Above the owner's minimum wall	Thickness record per TML	S	R	R				
6	Owner's decision on findings (repair, monitor or accept)	Owner integrity procedure	Decision recorded	Signed disposition	R	H	R				
7	Closed with new gaskets on every opened joint; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
8	Service leak test of opened joints on start-up	SW-TEST-2	No leak	Leak test record	W	W	—				
9	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
10	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 6"-P300-P-1012 — Internal inspection of horizontal drum D-101





BILL OF MATERIALS

1	GASKET 6" P300 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 6" P300	2
3	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300	4
4	FLANGE WELD-NECK 6" P300 RAISED FACE	2
5	PIPE 6" P300 BEVELLED ENDS	69'-11-1/16"
6	ELBOW 90° R = 5/3 OD 6" P300	5
7	TEE REDUCING 6" × 3/4" P300	1

WELDS 15 (2 FIELD) · SUPPORTS 3 · SPOOLS 3

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13 W14 W15F

6"-P300-P-1012

CLASS H300 — Hydrocarbon process
SERVICE P · 6" · P300 · DESIGN 300 psi
E-103-N4 → D-101-N1
PIPE 69'-11-1/16" · WEIGHT 1498 lb

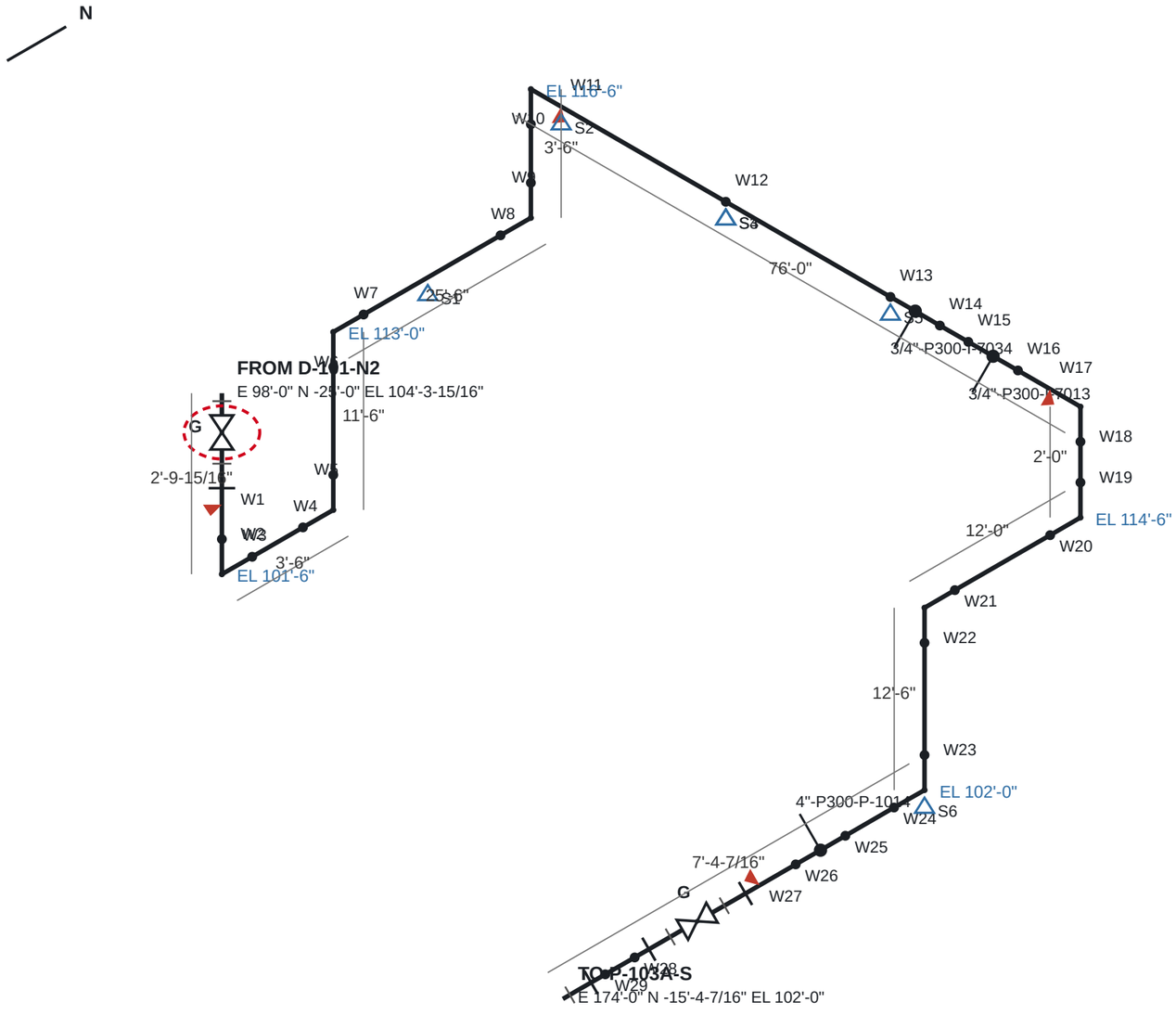
DWG ISO-6"-P300-P-1012 REV 0 WO-300-03

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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Isometric ISO-4"-P300-P-1013

ISOMETRIC — 4"-P300-P-1013 — Internal inspection of horizontal drum D-101



BILL OF MATERIALS

1	GASKET 4" P300 COMPRESSED SHEET	5
2	VALVE, GATE, WEDGE, OS&Y 4" P300	2
3	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	5
4	FLANGE WELD-NECK 4" P300 RAISED FACE	4
5	PIPE 4" P300 BEVELLED ENDS	140'-7-1/2"
6	ELBOW 90° R = 5/3 OD 4" P300	9
7	TEE REDUCING 4" × 3/4" P300	2
8	TEE EQUAL 4" P300	1

WELDS 29 (4 FIELD) · SUPPORTS 6 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11F W12 W13 W14 W15 W16 W17F W18 W19 W20
W21 W22 W23 W24 W25 W26 W27F W28 W29

4"-P300-P-1013

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
D-101-N2 → P-103A-S
PIPE 140'-7-1/2" · WEIGHT 1562 lb

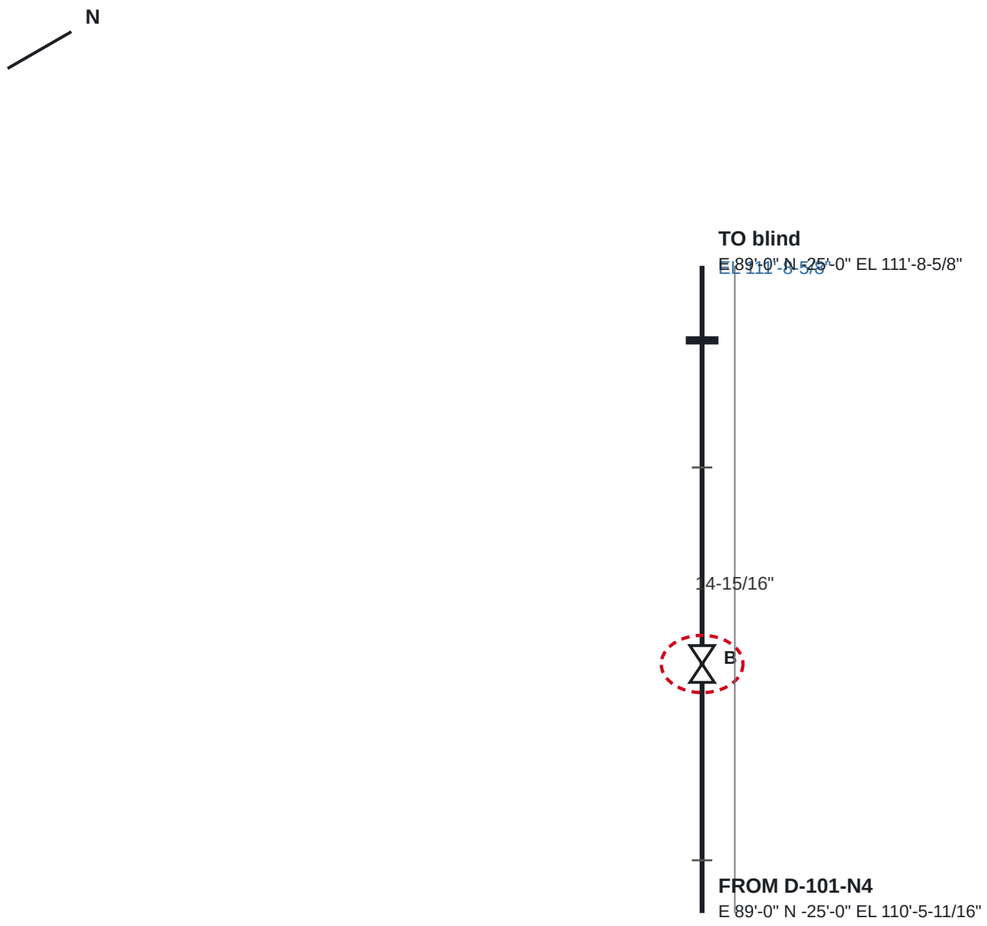
DWG ISO-4"-P300-P-1013 REV 0 WO-300-03

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Isometric ISO-1"-P150-V-6016

ISOMETRIC — 1"-P150-V-6016 — Internal inspection of horizontal drum D-101



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-V-6016

CLASS H150 — Hydrocarbon process
SERVICE V · 1" · P150 · DESIGN 150 psi
D-101-N4 → blind
PIPE 0.000" · WEIGHT 39 lb

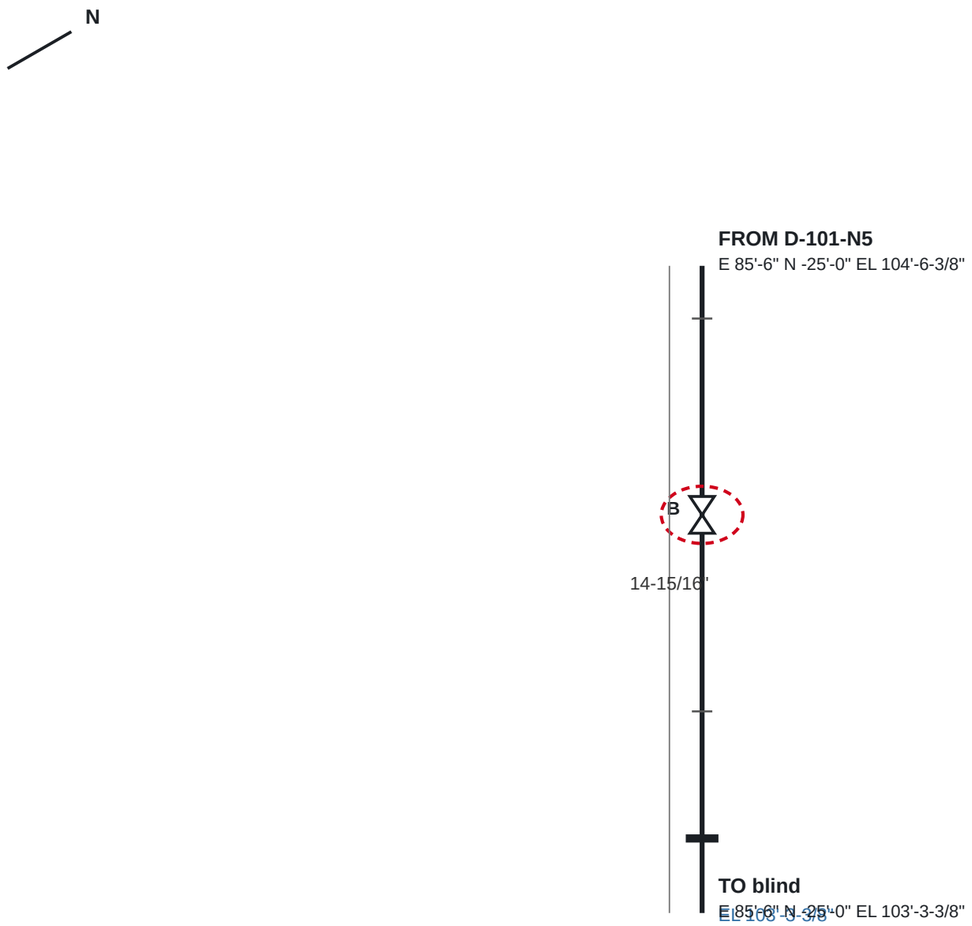
DWG ISO-1"-P150-V-6016 REV 0 WO-300-03

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DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

Isometric ISO-1"-P150-D-6017

ISOMETRIC — 1"-P150-D-6017 — Internal inspection of horizontal drum D-101



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-D-6017

CLASS H150 — Hydrocarbon process
SERVICE D · 1" · P150 · DESIGN 150 psi
D-101-N5 → blind
PIPE 0.000" · WEIGHT 39 lb

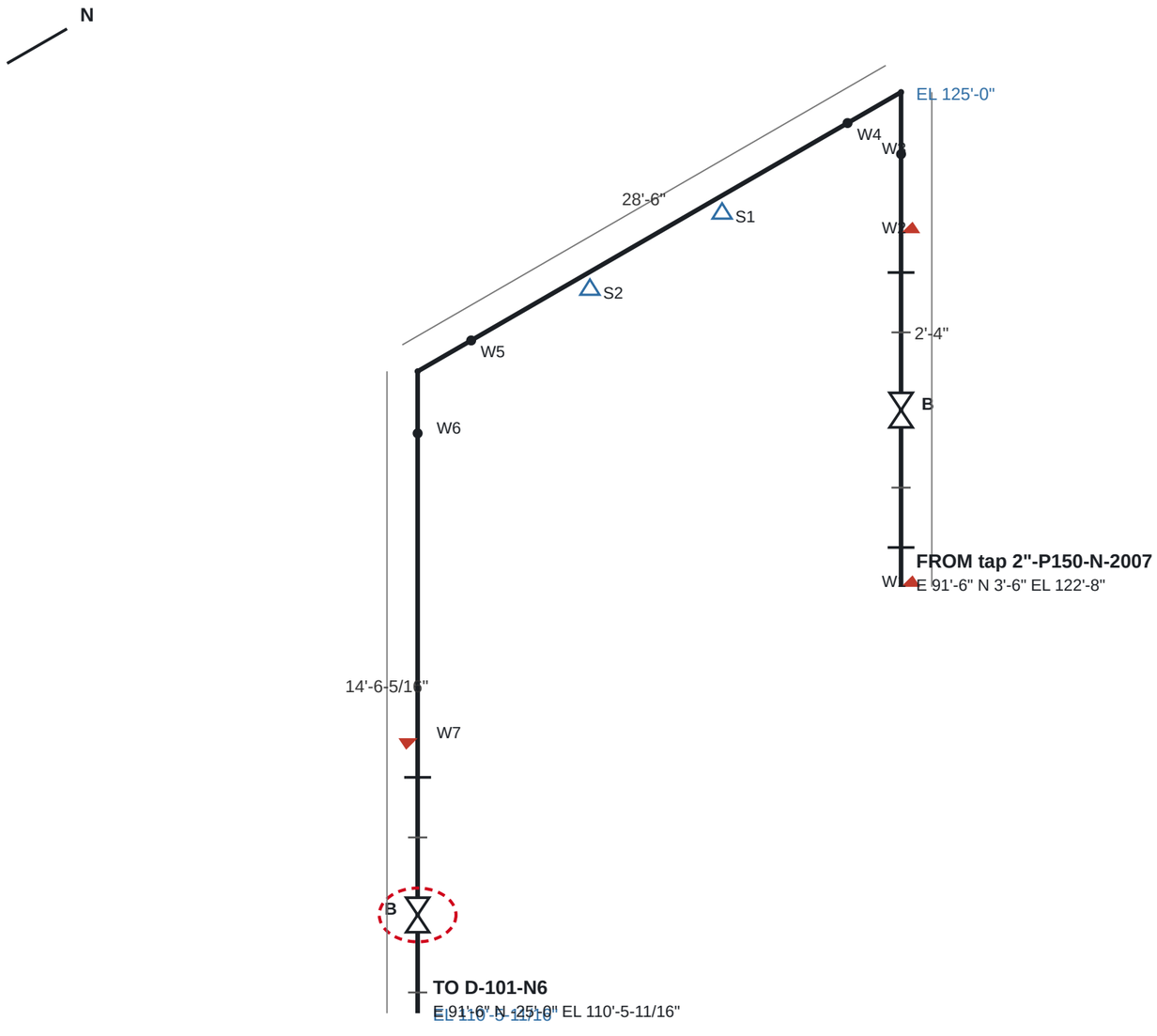
DWG ISO-1"-P150-D-6017 REV 0 WO-300-03

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

Isometric ISO-1"-P150-N-3014

ISOMETRIC — 1"-P150-N-3014 — Internal inspection of horizontal drum D-101



BILL OF MATERIALS

1	FLANGE WELD-NECK 1" P150 RAISED FACE	3
2	GASKET 1" P150 COMPRESSED SHEET	4
3	VALVE, BALL, FLOATING 1" P150	2
4	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	4
5	PIPE 1" P150 BEVELLED ENDS	41'-5-7/8"
6	ELBOW 90° R = 5/3 OD 1" P150	2

WELDS 7 (3 FIELD) · SUPPORTS 2 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7F

1"-P150-N-3014

CLASS G150 — Air and gases
SERVICE N · 1" · P150 · DESIGN 150 psi
tap 2"-P150-N-2007 → D-101-N6
PIPE 41'-5-7/8" · WEIGHT 121 lb

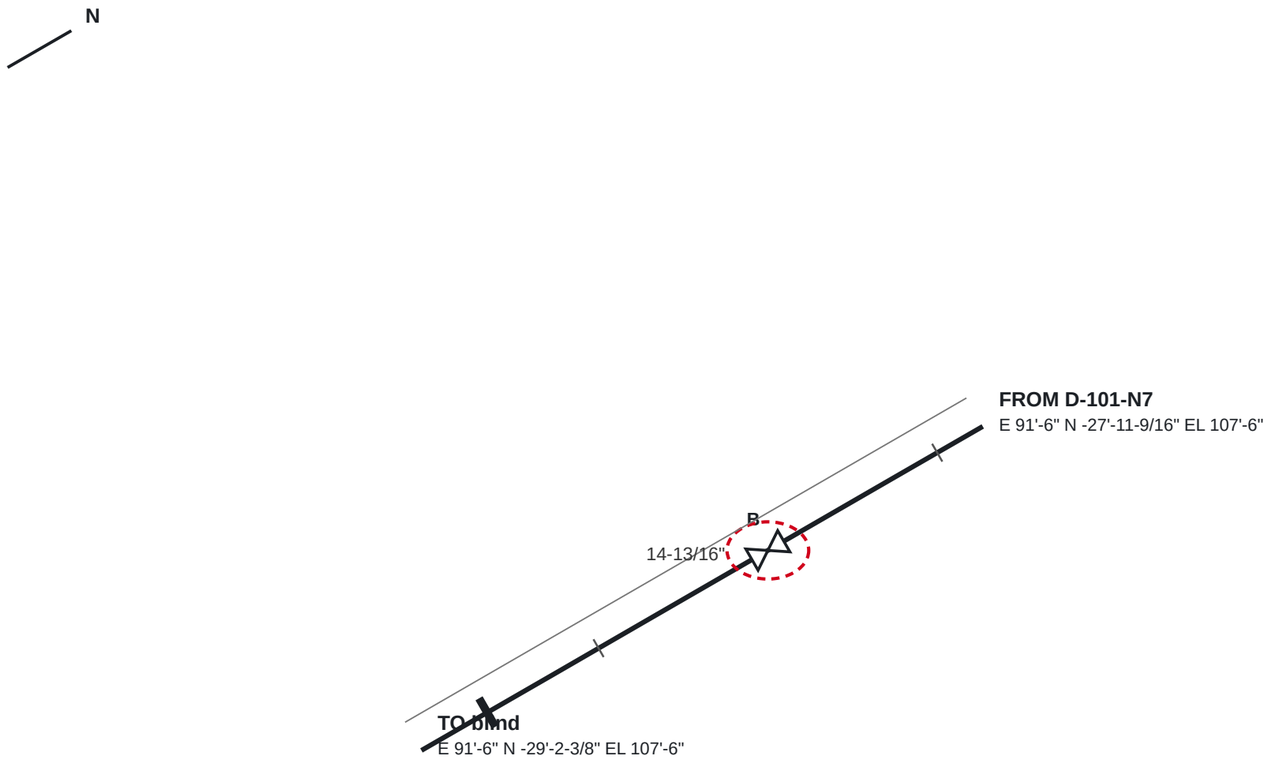
DWG ISO-1"-P150-N-3014 REV 0 WO-300-03

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Isometric ISO-3/4"-P300-I-6018

ISOMETRIC — 3/4"-P300-I-6018 — Internal inspection of horizontal drum D-101



BILL OF MATERIALS

1	GASKET 3/4" P300 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P300	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
4	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P300-I-6018

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
D-101-N7 → blind
PIPE 0.000" · WEIGHT 32 lb

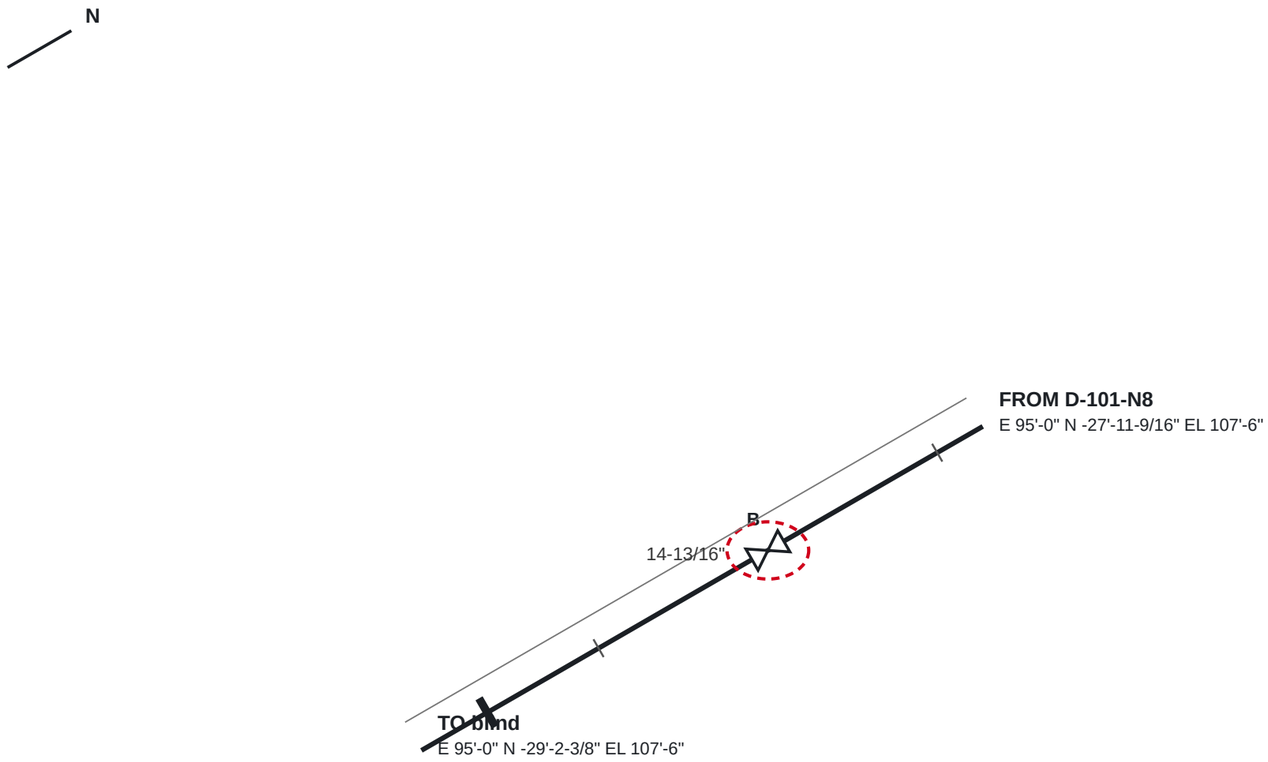
DWG ISO-3/4"-P300-I-6018 REV 0 WO-300-03

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

Isometric ISO-3/4"-P300-I-6019

ISOMETRIC — 3/4"-P300-I-6019 — Internal inspection of horizontal drum D-101



BILL OF MATERIALS

1	GASKET 3/4" P300 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P300	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
4	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P300-I-6019

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
D-101-N8 → blind
PIPE 0.000" · WEIGHT 32 lb

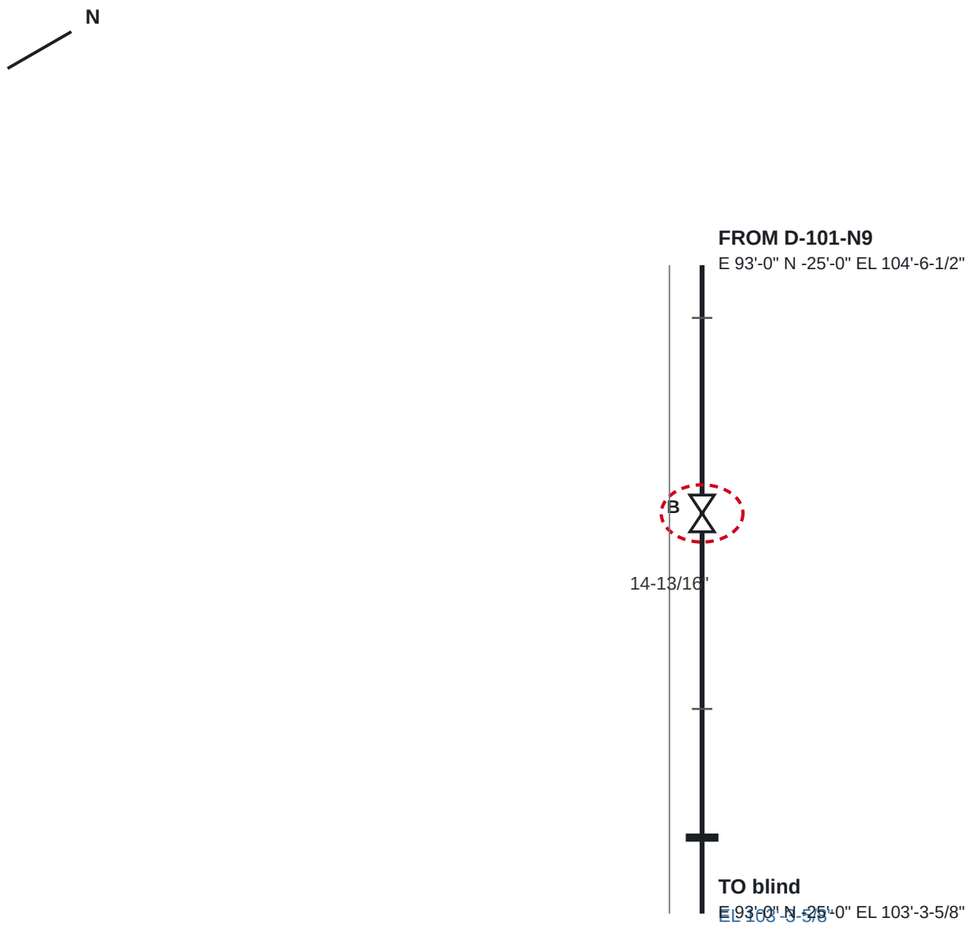
DWG ISO-3/4"-P300-I-6019 REV 0 WO-300-03

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

Isometric ISO-3/4"-P300-SP-6020

ISOMETRIC — 3/4"-P300-SP-6020 — Internal inspection of horizontal drum D-101



BILL OF MATERIALS

1	GASKET 3/4" P300 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P300	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
4	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P300-SP-6020

CLASS H300 — Hydrocarbon process
SERVICE SP · 0.75" · P300 · DESIGN 300 psi
D-101-N9 → blind
PIPE 0.000" · WEIGHT 32 lb

DWG ISO-3/4"-P300-SP-6020 REV 0 WO-300-03

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE